

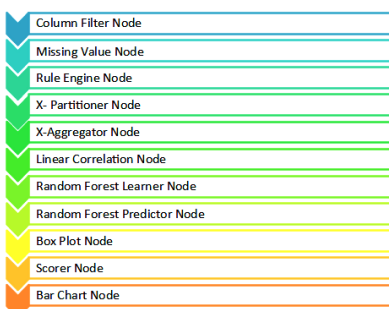
How to Make a Calories Burn Prediction on a Fitness Tracker Data Set

The KNIME Analytics Platform is open source software built for use in data science. Here's a short tutorial on how to use it to analyse a FitBit fitness tracker data set to predict the number of calories that will be burnt.



KNIME makes the building of data science workflows and reusable components accessible to everyone by being intuitive, open, and evolving continuously. This article outlines how to make a calorie burn prediction with a random forest predictor through the KNIME Analytics Platform, using a fitness tracker data set.

The KNIME nodes that will be needed for this are listed below:



Building the fitness tracker flow

1. Download the 'FitBit fitness tracker data' data set from <https://www.kaggle.com/arashnic/fitbit>.
2. This data set consists of 15 column attributes, namely: Id, Activity date, Total steps, Total distance, Logged activity distance, Very active distance, Moderately active distance, Light active distance, Sedentary active distance, Very active minutes, Fairly active minutes, Lightly active minutes, Sedentary minutes and Calories. All of the first 14 columns determine the individual's overall calories burnt in a day. With this data set, we can identify if an individual has a high, low, or good calorie burn for the day.
3. Drag and drop the .csv file in the KNIME platform to get the CSV Reader node.
4. Connect this with the Column filter and Missing Value node for preprocessing the data. In the Column filter, exclude the ActivityDate column, as the date doesn't matter for this prediction. In the Missing Value node, enter all the missing values as the mean of the entire column.
5. Connect the Missing Value node with the Rule Engine node. Here, give the rules as follows:
 - If Calories > 2500 => High calories burn
 - If Calories < 2000 => Low calories burn
 - If Calories >= 2000 and Calories <= 2500 => Good calories burn.

Append this data into a new column named 'Calories burn'.